

PROJECT SYNOPSIS

FOCP-II - BTech 2nd Semester

01. Project Title

Title: _Darwin Evolution Simulation_

02. Team Details

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Section / Batch	F
Semester	2nd Semester (FOCP-II)
Submission Date	30-03-2026

03. Problem Statement

Clearly describe the problem your project is solving. (3–5 lines)

Can complex behaviour emerge from simple, randomized beginnings?

This project replaces "if-else" AI with Automated Strategy Discovery, using environmental pressures (hunger, predation) to evolve optimal survival traits.

04. Objective of the Project

Deadline 6-04-2026

What is the goal of your program? List 3–5 clear objectives.

- **Simulate Natural Selection:** To create a digital environment where entities (agents) compete for resources, demonstrating the "survival of the fittest" principle.
- **Model Genetic Inheritance:** To implement a system where traits (speed, size, sensory range) are passed from parents to offspring with slight variations.
- **Analyze Population Dynamics:** To track and visualize how a population changes over time in response to environmental pressures like food scarcity or predators.
- **Demonstrate Mutation:** To introduce random genetic mutations to show how new, potentially beneficial traits can emerge within a closed ecosystem.

05. Features / Functionalities

Deadline 6-04-2026

List all the things your program can do. Examples shown below:

- **Entity Creation (Add Record):** Initialize a starting population with randomized genetic traits such as health, energy levels, and movement speed.
- **Real-time Environment Interaction:** Logic for entities to "eat" food to gain energy and "reproduce" once an energy threshold is met.
- **Death & Cleanup (Delete Record):** Automated removal of entities from the simulation once their health or energy reaches zero.
- **Trait Evolution Tracking (Display / List all data):** A dashboard or console output that displays the average statistics of the current generation vs. previous generations.

06. Concepts Used (C++)

Deadline 6-04-2026

Tick / list all C++ concepts applied in your project:

- OOP: Inheritance and Polymorphism for organisms and hazards.
- STL: `std::vector` for populations; `std::priority_queue` for events.
- Memory: `std::unique_ptr` for automated birth/death lifecycle management.
- File I/O: Saving/loading successful Genomes via `.txt` or `.json`.
- Operators: Overloading `<` and `==` for fitness ranking and genomic comparisons.
- Randomness: `<random>` header for mutation and survival probability.

07. Program Flow / Logic

Deadline 20-04-2026

Explain how your program works step by step. Example flow shown below:

1. **Initialization:** The program starts by prompting the user to input simulation parameters (e.g., initial population size, mutation rate, and food abundance).
2. **Entity Spawning:** A collection of "Creature" objects/structs is created, each initialized with a random "DNA" string (representing traits like speed, size, and metabolism).
3. **Environment Generation:** The simulation grid is populated with food resources at random coordinates.
4. **The Simulation Loop (Main Logic):**
 - **Perception:** Each entity "searches" the environment for the nearest food source using distance formulas.
 - **Movement:** Entities move toward food, consuming energy based on their speed and size traits.
 - **Metabolism:** If an entity's energy reaches zero before finding food, it is flagged as "dead" and removed from the record.
 - **Reproduction:** Entities that reach an energy threshold create a "clone" with a probability of **Genetic Mutation** (slight random variation in trait values).
5. **Data Logging:** At the end of each "Generation" or time step, the program calculates and stores population averages (e.g., "Average Speed of Survivors").

08. Input / Output Specifications

Deadline 20-04-2026

List every input your program accepts and every output it produces:

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#	Input Description	Expected Output	Data Type / Format
1	Initial Population Count	Number of active entities spawned in the simulation grid.	<code>int</code> (e.g., 50)
2	Mutation Rate	Probability of a trait changing during reproduction.	<code>float</code> (e.g., 0.05 for 5%)
3	Search Query (Entity ID)	Detailed stats of a specific entity (Speed, Energy, Age).	<code>int</code> or <code>string</code>
4	Simulation Commands	Execution of actions like "Start", "Pause", or "Generate Report".	<code>char</code> or <code>int</code> (Menu Selection)

09. Challenges Faced

Deadline 30-04-2026

Write 2–4 difficulties you encountered during development.

- **Memory Management:** Managing a dynamic population where entities are constantly being born (created) and dying (deleted) without causing memory leaks or segmentation faults.
- **Balancing Variables:** Fine-tuning the mathematical relationship between speed, size, and energy consumption so that the simulation doesn't end in immediate mass extinction or infinite population growth.
- **Search Efficiency:** Implementing an efficient way to search through a large list of active entities to display specific "genetic" data in real-time during the simulation.
- **Logic Complexity:** Debugging the reproduction logic to ensure that mutations were truly random and properly inherited rather than just resetting to default values.

10. Learning Outcome

Deadline 30-04-2026

What new skills or concepts did you learn from this project?

- **Dynamic Data Structures:** Gained hands-on experience using arrays or linked lists to manage a fluctuating set of data records (the creature population).
- **Algorithmic Thinking:** Developed a deeper understanding of how to translate complex biological theories (Darwinism) into logical code and mathematical formulas.
- **Modular Programming:** Learned the importance of writing clean, reusable functions for tasks like "Calculate Distance," "Update Health," and "Apply Mutation."
- **Problem-Solving:** Improved debugging skills by identifying and fixing logic errors in a real-time looping environment where multiple variables interact simultaneously.

11. Future Scope

Deadline 30-04-2026

What features or improvements could be added in the future?

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- Neural Integration: Evolving brain weights instead of basic logic.
 - Biomes: Varying terrain with unique survival trade-offs.
 - Speciation: Tracking reproductive isolation between diverging groups.
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